



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)
ALLEN COMMON TEST
12-09-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	B	A	A	A	A
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,D	A,C	B,C,D	A,B,C	A,B	B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	9.85 to 10.05	1.55 to 1.60	0.33	5.00	0.50	7.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	B	C	C	B	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C,D	A,B,C	B,D	B,D	B,C,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.76	2.00	18.00	3.00	0.00	0.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A	D	B	D	B	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,D	A,C	A,C,D	A,B,D	B,C,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	6.00	9.00	2.00	6.00	201.00	0.30

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

$$M = \rho_0 \left(\frac{4}{3} \pi R^3 \right)$$

$$g = \frac{GM}{R^2} \text{ (at the surface)}$$

$$\text{Now } R \longrightarrow 2R$$

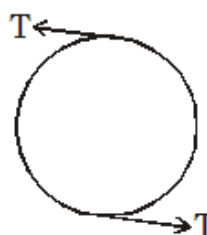
and M remains same

so at $r = R$

$$g' = \frac{GMr}{(2R)^3}$$

$$g' = \frac{GMR}{8R^3} = \frac{GM}{8R^2} = \frac{g}{8}$$

2. Ans (B)



$$\text{If } a_c = g \quad \text{then } \omega^2 R = g \quad \omega = \sqrt{\frac{g}{R}}$$

$$\tau = 2TR = I\alpha \quad 2TR = (MR^2)\alpha$$

$$\alpha = \frac{2T}{MR} \quad \omega_F = \omega_i + \alpha t$$

$$\sqrt{\frac{g}{R}} = \frac{2T}{MR} t \Rightarrow t = \frac{MR}{2T} \sqrt{\frac{g}{R}} = \frac{M}{2T} \sqrt{Rg}$$

$$\text{so time} = \frac{M}{2T} \sqrt{Rg}$$

3. Ans (A)

$\theta \rightarrow$ small

elongation in spring = $\ell\theta$

force = $-k\ell\theta$

$\tau = fd = -k\ell^2\theta$

$\tau = I\alpha$

so $I\alpha = -k\ell^2\theta$

$$\alpha = -\frac{k\ell^2\theta}{\frac{M\ell^2}{3}}$$

$$\omega = \sqrt{\frac{3k}{M}}$$

$$\alpha = -\frac{3k}{M}\theta$$

$\theta \rightarrow$ small

elongation = $\frac{\ell}{2}\theta$

force = $-k\frac{\ell}{2}\theta$

torque = $\frac{\ell}{2}f = -k\frac{\ell^2}{4}\theta$

$$\omega' = \sqrt{\frac{3k}{4M}}$$

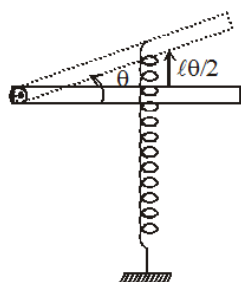
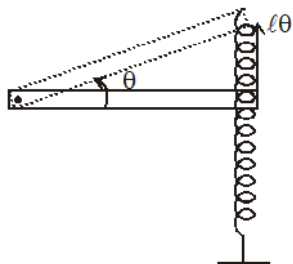
$I\alpha = -k\frac{\ell^2}{4}\theta$

$$f' = \frac{\omega}{2\pi} = \frac{1}{2} \left(\frac{1}{2\pi} \sqrt{\frac{3k}{M}} \right)$$

$$\alpha = -\frac{k\ell^2}{4\frac{M\ell^2}{3}}\theta$$

$$f' = \frac{f}{2}$$

$$\alpha = -\frac{3k}{4M}\theta$$



4. Ans (A)

$$V_{AVG} = \frac{\sqrt{\frac{m}{k}} V_0 \sin\left(\sqrt{\frac{k}{m}} \frac{T}{3}\right)}{T/3}$$

$$\text{But } T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$

$$\text{so } V_{AVG} = \frac{3V_0}{2\pi} \sin\left(\frac{2\pi}{3}\right)$$

(not required : $V_{AVG} = 0.4 V_0$)

5. Ans (A)

$$q_i = q_0 \left(1 - \frac{1}{K}\right)$$

$$= 20 \times 150 \text{ pF} \left(1 - \frac{3}{5}\right) = 1.2 \text{ nC}$$

6. Ans (A)

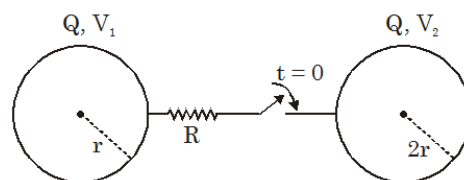
$$\text{At } X = \frac{A}{2}$$

$$a = \frac{\omega^2 A}{2} = \frac{a_0}{2}$$

$$\text{Also } t = \frac{T}{12} \text{ (calculated earlier)}$$

SECTION-I (ii)

7. Ans (A,D)

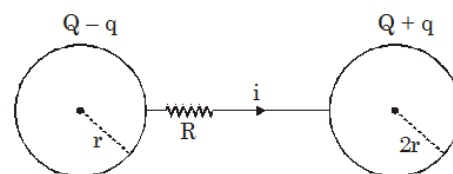


$$V_1 = \frac{kQ}{r}, V_2 = \frac{kQ}{2r}$$

At $t = 0$

$$i_0 = \frac{V_1 - V_2}{R} = \frac{kQ}{2rR} = \frac{Q}{8\pi\epsilon_0 Rr}$$

At $t = t$



$$V_1 = \frac{k(Q-q)}{r}, V_2 = \frac{k(Q+q)}{2r}$$

$$\Delta V = V_1 - V_2 = \frac{k(Q-3q)}{2r}$$

$$i = \frac{\Delta V}{R} = \frac{k(Q-3q)}{2Rr} = \frac{i_0}{2}$$

$$q = \frac{Q}{6}$$

$$i = \frac{k(Q-3q)}{2Rr} = \frac{dq}{dt}$$

$$\frac{k}{2Rr} \int_0^t dt = \int_0^{\frac{Q}{6}} \frac{dq}{Q-3q}$$

$$\frac{k}{2Rr} t = -\frac{1}{3} (\ln(Q-3q))_0^{\frac{Q}{6}}$$

$$t = \frac{8\pi\epsilon_0 Rr \ln 2}{3}$$

8. **Ans (A,C)**

$$U = \frac{Q^2 d}{2\epsilon_0 A} \quad V = \frac{Qd}{\epsilon_0 A}$$

9. **Ans (B,C,D)**

$$(a) V = \frac{q}{4\pi\epsilon_0 x} = 600$$

$$E = \frac{q}{4\pi\epsilon_0 x^2} = 200$$

Dividing, $x = 3m$

(b) from eqn. (i),

$$q = \frac{600 \times 3}{9 \times 10^9} = 20 \times 10^{-6} C$$

$$(c) V' = \frac{q}{4\pi\epsilon_0 x} = \frac{20 \times 10^{-6} \times 9 \times 10^9}{9} = 200V$$

$$(d) W = q(V - V') = 10^{-6} (600 - 200) = 4 \times 10^{-4} J$$

10. **Ans (A,B,C)**

$$E = K_0 = U_0 = \frac{1}{2} kA^2 = \frac{1}{2} k(0.1)^2$$

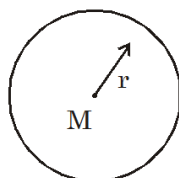
$$\text{At } x = 0.06, \quad U = \frac{1}{2} k(0.06)^2$$

$$K = \frac{1}{2} k(0.1)^2 - \frac{1}{2} k(0.06)^2 = 0.64 E.$$

$$x = 0.05, \quad U = \frac{1}{2} k(0.05)^2$$

$$\therefore K = \frac{1}{2} k(0.1)^2 - \frac{1}{2} k(0.05)^2 = 0.75 E$$

11. **Ans (A,B)**



$$\text{when } r < R \quad g = \frac{GM r}{R^3}$$

$$\text{when } r > R \quad g = \frac{GM}{r^2} \quad \text{so for } r_1 < R \text{ and } r_2 < R$$

$$F_1 = \frac{GM r_1}{R^3}; \quad F_2 = \frac{GM r_2}{R^3}$$

$$\frac{F_1}{F_2} = \frac{r_1}{r_2}$$

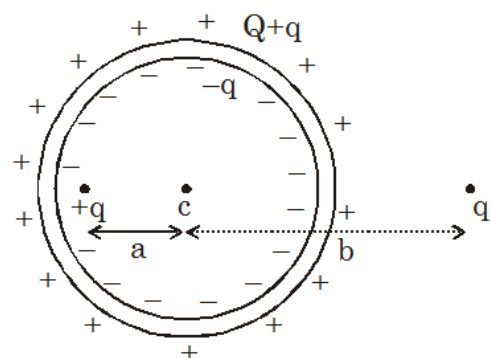
and

for $r_1 > R$ and $r_2 > R$

$$F_1 = \frac{GM}{r_1^2}; \quad F_2 = \frac{GM}{r_2^2}$$

$$\frac{F_1}{F_2} = \frac{r_2^2}{r_1^2} = \left(\frac{r_2}{r_1}\right)^2$$

12. **Ans (B,D)**



EF due to $(Q + q)$ & q_1 i.e. outer charges is zero inside the conductor.

$$\Rightarrow \vec{E}_{Q+q} + \vec{E}_{q_1} = 0$$

$$\Rightarrow |E_{Q+q}| = |E_{q_1}| = \frac{kq_1}{b^2}$$

$$V_C = V_{+q} + V_{-q} + V_{Q+q} + V_{q_1}$$

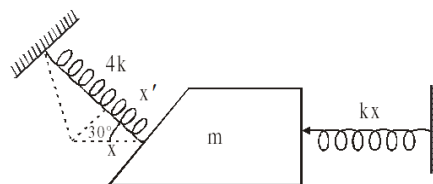
$$= \frac{kq}{a} + \frac{k(-q)}{R} + \frac{k(Q+q)}{R} + \frac{kq_1}{b}$$

$$V_C = \frac{kq}{a} + \frac{kQ}{R} + \frac{kq_1}{b}$$

SECTION-II

1. **Ans (9.85 to 10.05)**

When block is displaced slightly by x slightly towards right then $x' = x \cos 30^\circ$.



The restoring force

$$F = -(kx + 4kx' \cos 30^\circ)$$

$$= -(kx + 4k(x \cos 30^\circ) \cos 30^\circ) = -[kx + 3kx] = -4kx$$

$$a = \frac{F}{m} = \frac{4k}{m} (-x)$$

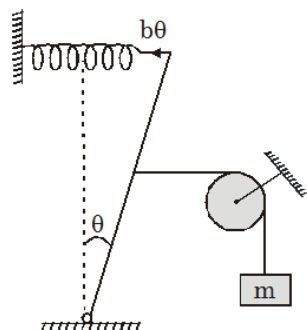
On comparing with $a = -\omega^2 x$, we get

$$\omega = \sqrt{\frac{4k}{m}} \quad \text{and} \quad T = 2\pi \sqrt{\frac{m}{4k}} = \pi \sqrt{\frac{m}{k}}$$

$$= \pi \sqrt{\frac{22 \times 22}{7 \times 7}} = \frac{22}{7} \times \frac{22}{7} = 9.87$$

2. Ans (1.55 to 1.60)

In this device, the restoring torque is constituted by spring force ($F = kb\theta$). So for small θ



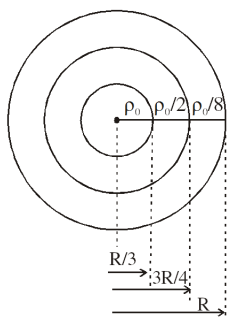
$$\tau_{\text{rest}} = -(kb\theta) \times b$$

$$\text{and } \alpha = \frac{\tau_{\text{rest}}}{I} = \frac{kb^2}{ma^2}(-\theta)$$

on comparing with, $\alpha = -\omega^2\theta$, we get

$$\omega = \sqrt{\frac{kb^2}{ma^2}} \text{ or } T = 2\pi\sqrt{\frac{ma^2}{kb^2}}.$$

3. Ans (0.33)



The gravitational field at a distance r due to a spherical body of mass M is given by

$$E = \frac{GM}{r^2}$$

$$(i) \text{ For } r = \frac{R}{4}, \text{ density } \rho = \rho_0$$

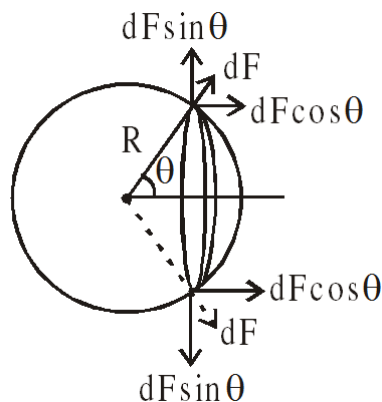
The mass of the sphere of radius $R/4$

$$M = \frac{4}{3}\pi\left(\frac{R}{4}\right)^3 \rho_0 = \pi R^3 \rho_0 / 48$$

Gravitational field at a distance $R/4$ from the centre of sphere

$$\therefore E_1 = \frac{GM}{r^2} = \frac{G \times [\pi R^3 \rho_0 / 48]}{(R/4)^2} = 0.33\pi G R \rho_0$$

4. Ans (5.00)



Take an elementary area dA containing a charge dq on the elementary ring of radius r .

The force acting on dq is

$$dF = \frac{\sigma^2}{2\epsilon} dA$$

$$dA = 2\pi (R \sin\theta) R d\theta$$

The net force on the ring is

$$(dF_{\text{net}})_{\text{ring}} = dF \cos\theta$$

$$= \left(\frac{\sigma^2}{2\epsilon_0}\right) dA \cos\theta$$

$$= \left(\frac{\sigma^2}{2\epsilon_0}\right) (2\pi R \sin\theta) (R d\theta) \cos\theta$$

$$\therefore F_{\text{hemisphere}} = \int (dF_{\text{net}})_{\text{ring}}$$

$$= \frac{\sigma^2 \pi R^2}{\epsilon_0} \int_0^{\pi/2} (\sin\theta \cos\theta) d\theta,$$

$$\text{where, } \sigma = \frac{Q}{4\pi R^2}$$

$$= \frac{Q^2}{32\pi\epsilon_0 R^2} = \frac{Q^2}{2^5\pi\epsilon_0 R^2}$$

so, $n = 5$

5. **Ans (0.50)**

Without cavity: The potential at A and electric field at C due to the whole solid sphere are given by

$$V_A' = \frac{3}{2} \times \left(\frac{1}{4\pi\epsilon_0} \right) \frac{(4/3)\pi R^3 \rho}{R} = \frac{\rho R^2}{2\epsilon_0}$$

$$E_C' = \frac{\rho}{3\epsilon_0} AC$$

The potential at A electric field at C due to cavity alone are given by

$$V_A'' = \left(\frac{1}{4\pi\epsilon_0} \right) \frac{(4/3)\pi (R/2)^3 \rho}{(R/2)} = \frac{\rho R^2}{12\epsilon_0}$$

$$E_C'' = \frac{\rho}{3\epsilon_0} BC$$

The net potential at A and electric field at C due to the given sphere with cavity is the difference the above two cases.

Therefore,

$$V_A = V_A' - V_A'' = \frac{\rho R^2}{2\epsilon_0} - \frac{\rho R^2}{12\epsilon_0} = \frac{5\rho R^2}{12\epsilon_0}$$

$$\Rightarrow x = 12$$

$$\vec{E}_C = \vec{E}_C' + (-\vec{E}_C'')$$

$$= \frac{\rho}{3\epsilon_0} (\vec{AC} - \vec{BC}) = \frac{\rho}{3\epsilon_0} (\vec{AB})$$

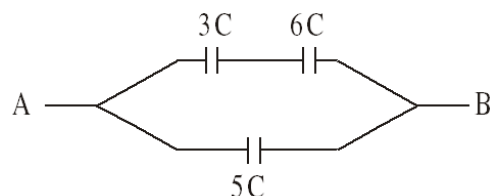
$$|\vec{E}_C| = \frac{\rho}{3\epsilon_0} \left(\frac{R}{2} \right) = \frac{\rho R}{6\epsilon_0}$$

$$\Rightarrow x = 6$$

$$\frac{x}{y} = \frac{12}{6} = 2$$

6. **Ans (7.00)**

The equivalent circuit is shown in figure.



Therefore, the equivalent capacity between A and B is given by

$$C_{eq} = \frac{3C \times 6C}{3C + 6C} + 5C = 2C + 5C = 7C$$

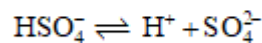
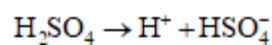
$$\text{where } C = \epsilon_0 A/d$$

$$\therefore n = 7$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. **Ans (B)**



$$\text{total concentration} = 1 - \alpha + 1 + \alpha + \alpha = 2 + \alpha$$

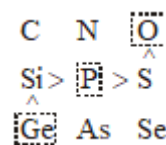
$$\pi = CTRT \pi (2 + \pi) \times 0.082 \times 500 = 90.2$$

$$\Rightarrow \pi = 0.2$$

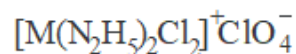
$$\text{Now } K_{a2} = \frac{[H^+][SO_4^{2-}]}{[HSO_4^-]} = \frac{(1 + 0.2)(0.2)}{1 - 0.2}$$

$$\Rightarrow K_{a2} = \frac{1.2 \times 0.2}{0.8} = 0.3$$

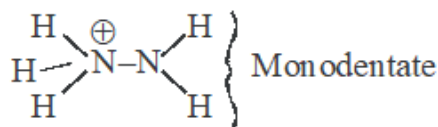
3. **Ans (C)**



4. **Ans (C)**



Hydrazinium ion



$$\therefore \text{Co. No.} = 4$$

$$x + 2(+1) + 2(-1) = +1$$

$$x = +1 \text{ (ox number of central atom)}$$

SECTION-I (ii)

8. **Ans (A,B,C)**

The given binary solution may not be azeotrope.

9. **Ans (B,D)**

(I) incorrect M.O.

(II) Hund's rule violation

(III) ground state configuration

(IV) $3e^-$ in each orbital

10. **Ans (B,D)**

$[\text{Zn}(\text{NH}_3)_4]^{2+}$: diamagnetic, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$: paramagnetic

Option (B) $\Rightarrow$ Both are paramagnetic
 $\Rightarrow$ Both exhibit geometrical isomerism

$[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-} \Rightarrow$ diamagnetic, $[\text{Fe}^{+3}(\text{EDTA})]^{-1} \Rightarrow$ paramagnetic

Option (D) $\Rightarrow$ Both are diamagnetic
 $\Rightarrow$ Both exhibit linkage isomerism

SECTION-II

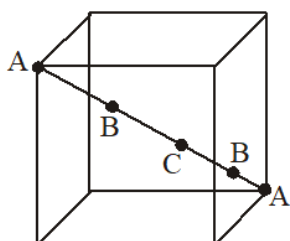
1. **Ans (0.76)**

Length of body diagonal covered up = x

$$x = 2r_A + 4r_B + 2r_C$$

$$x = 2r_A + 4 \times 0.225 r_A + 2 \times 0.414 r_A$$

$$\Rightarrow x = 3.728 r_A \quad \dots(1)$$



$$r_B = 0.225 r_A$$

$$r_C = 0.414 r_A$$

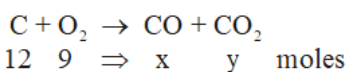
$$\text{and } \sqrt{2}a = 4r_A$$

$$\text{length of body diagonal} = \sqrt{3}a = \frac{4\sqrt{3}r_A}{\sqrt{2}} \dots(2)$$

$$\therefore \text{fraction of body diagonal covered} = \frac{x}{\sqrt{3}a}$$

$$= \frac{3.728}{4 \times 1.732} \times 1.414 \simeq 0.7609 \simeq 0.76$$

2. **Ans (2.00)**



$$\Rightarrow 12 = x + y$$

$$9 \times 2 = x + 2y$$

$$\Rightarrow y = 6 \text{ \& } x = 6$$

$$M_{\text{av.}} = \frac{6 \times 28 + 6 \times 44}{6 + 6} = 36$$

$$d = \frac{PM}{RT} = \frac{1.642 \times 36}{0.0821 \times 360}$$

$$= 2 \text{ g/lit}$$

3. **Ans (18.00)**

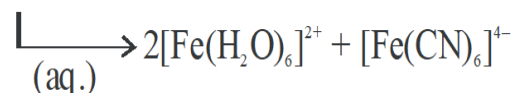
$$x = 3 \Rightarrow \text{CO}_2, \text{NO}_2, \dot{\text{C}}\text{H}_3$$

$$y = 3 \Rightarrow \text{NO}_2, \dot{\text{C}}\text{F}_3, \text{ClO}_3$$

$$z = 2 \Rightarrow \text{CH}_4, \dot{\text{C}}\text{F}_3$$

4. **Ans (3.00)**

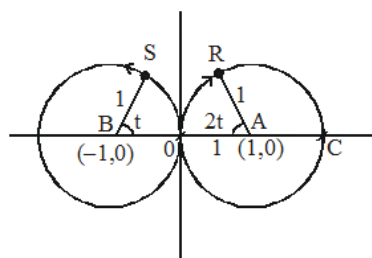
$[\text{Fe}(\text{H}_2\text{O})_6]_2[\text{Fe}(\text{CN})_6] \equiv$ complex salt



PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A)**



Let at time 't' Ram is at R and shyam is at S

$$\widehat{\text{OR}} = 2t \Rightarrow \angle \text{OAR} = 2t$$

$$\widehat{\text{OS}} = t \Rightarrow \angle \text{OBS} = t$$

coordinates of R $(1 - \cos 2t, \sin 2t)$

and coordinates of S $(\cos t - 1, \sin t)$

$$RS = \sqrt{(\cos t + \cos 2t - 2)^2 + (\sin t - \sin 2t)^2}$$

on differentiating w.r.t. 't' and putting

$$2t = \pi \text{ i.e. } t = \frac{\pi}{2}$$

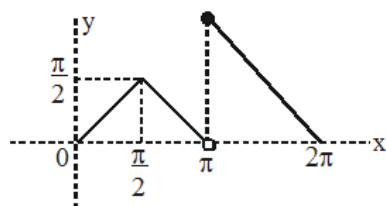
$$\Rightarrow RS = \sqrt{\frac{5}{2}} \text{ at point C.}$$

2. **Ans (D)**

$$f(r) = \lim_{x \rightarrow \infty} e^{\frac{-2rx^2}{2x^2+3r}} = e^{-r}$$

$$\sum_{r=1}^{\infty} f(r) = \frac{1/e}{1 - \frac{1}{e}} = \frac{1}{e-1}$$

3. Ans (B)



Local maxima at $x = \frac{\pi}{2} \& \pi$

4. Ans (D)

$$(\cot x)(\cot^3 x) = \cot^4 x = \cot^4 \left(\frac{\pi}{10} \right) > 1$$

$$f(x) = \pi + \tan^{-1} \left(\frac{\cot x + \cot^3 x}{1 - (\cot x)(\cot^3 x)} \right) + \tan^{-1} \left(\frac{\tan x}{1 - \tan^2 x} \right)$$

$$= \pi + \tan^{-1} \left(\frac{\tan x}{\tan^2 x - 1} \right) + \tan^{-1} \left(\frac{\tan x}{1 - \tan^2 x} \right)$$

$$= \pi$$

5. Ans (B)

$$f(x) = \frac{ax^4}{4} + \frac{bx^3}{3} + \frac{cx^2}{2} + dx$$

$$f(-1) = \frac{a}{4} - \frac{b}{3} + \frac{c}{2} - d = 3a - 4b + 6c - 12d = 0$$

$$f(0) = 0$$

$\therefore \exists$ exist at least one root in $[-1, 0)$ acc. to Rolle's

6. Ans (B)

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \left(\frac{\frac{r-\sqrt{r^2-1}}{\sqrt{r^2+r}}}{1 + \sqrt{\frac{r^2-r}{r^2+r}}} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \left(\frac{\sqrt{\frac{r}{r+1}} - \sqrt{\frac{r-1}{r}}}{1 + \sqrt{\frac{r}{r+1} \times \frac{r-1}{r}}} \right)$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\tan^{-1} \left(\sqrt{\frac{r}{r+1}} \right) - \tan^{-1} \left(\sqrt{\frac{r-1}{r}} \right) \right)$$

$$= \lim_{n \rightarrow \infty} \tan^{-1} \sqrt{\frac{n}{n+1}} = \frac{\pi}{4}$$

SECTION-I (ii)

7. Ans (A,D)

$$f(x) = a^2 x^3 + abx^2 - a^2 x - ab$$

$$f'(x) = 3a^2 x^2 + 2abx - a^2$$

$$f'(0) = -a^2 < 0$$

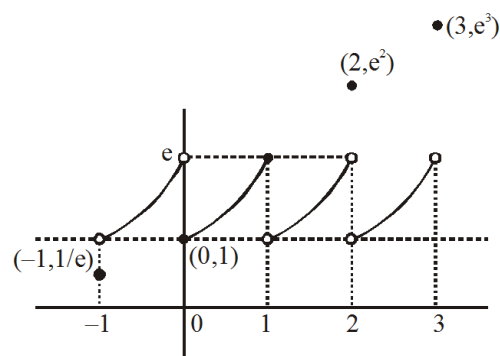
$\therefore x_1 < 0 \& x_2 > 0$

Now $x_1 + x_2 = -\frac{2ab}{3a^2} = -\frac{2}{3} \frac{b}{a} > 0 \Rightarrow \frac{b}{a} < 0$

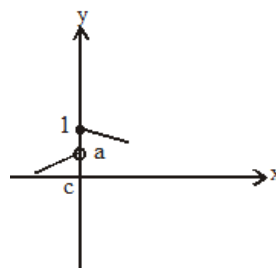
$$f'(1) = 2a^2 + 2ab < 0 \text{ for some } m \in \mathbb{R}.$$

8. Ans (A,C)

Graph of $f(x)$ will be



9. Ans (A,C,D)



$$x > 0 \quad f'(x) = 3x^2 - 12x = 3x(x - 4)$$

$f(x)$ decreases in $(0, 4)$

$$x < 0 \quad f(x) = 2x + 2 \Rightarrow f(x) \text{ increases in } (-1, 0)$$

$a \leq 1 \quad x = 0$ is point of maxima

$a > 1 \quad x = 0$ is point of decreasing

10. Ans (A,B,D)

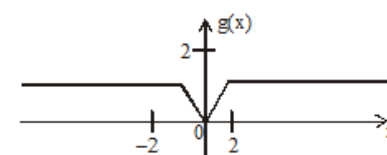
$$f'(x) = 3(x^2 - 2x + 2) - 2\sqrt{2} \cos x + \sin x$$

$$f'(x) = 3((x-1)^2 + 1) + (\sin x - 2\sqrt{2} \cos x) > 0 \Rightarrow f(x) \text{ is } \uparrow$$

$$f''(x) = 6(x-1) + \cos x + 2\sqrt{2} \sin x$$

$$f''(x) = 6 - \sin x + 2\sqrt{2} \cos x > 0 \Rightarrow f''(x) \text{ is } \uparrow$$

11. Ans (B,C,D)



$$f(x) = [2 \cot^{-1} x]. \text{ Range of } f(x) \text{ is } \{0, 1, 2, 3, 4, 5, 6\}$$

$f(x)$ is discontinuous at points where $2 \cot^{-1} x = 1, 2, 3, 4, 5, 6$

$$g(x) = \min(|x|, 2)$$

Range of $g(x)$ is $[0, 2]$

$g(x)$ is non-differentiable at $x = -2, 0, 2$

12. Ans (A,B,C,D)

$$f'(x) = e^x + 3x^2 - 2x + 5 > 0 \Rightarrow f(x) \uparrow$$

Hence one-one

Range of $f(x)$ is $\mathbb{R}$

$$f'(0) = 6 \Rightarrow \text{Equation of tangent at } (0, 4) \text{ is } y - 4$$

$$= 6(x - 0)$$

$$\Rightarrow y = 6x + 4$$

$$\text{Subnormal} = \left| y \frac{dy}{dx} \right| = 4 \times 6 = 24$$

SECTION-II

1. Ans (6.00)

$$f(0) = \sin 1 + -1$$

$$f(0^+) = \sin \lambda + 1$$

$$f(0) > f(0^+) \Rightarrow \sin \lambda < \sin 1$$

$$\Rightarrow \lambda = 0, 1, 3, 4, 5, 6$$

2. Ans (9.00)

$$x = \sin^{-1} \alpha + \frac{\pi \alpha}{2}, \alpha \in [-1, 1]$$

$$\Rightarrow \text{Max. value of } x = \sin^{-1} 1 + \frac{\pi}{2} = \pi \text{ at } \alpha = 1$$

$$\text{Min. value of } x = \sin^{-1}(-1) - \frac{\pi}{2} = -\pi \text{ at } \alpha = -1$$

$$\text{Now } y = \cos^{-1}(\beta - 2) + \frac{3\pi}{2\beta}$$

$$\text{Domain : } \beta \in [1, 3]$$

$$\text{Max. value of } y = \cos^{-1}(-1) + \frac{3\pi}{2} = \frac{5\pi}{2} \text{ at } \beta = 1$$

$$\text{Min. value of } y = \cos^{-1} 1 + \frac{\pi}{2} = \frac{\pi}{2} \text{ at } \beta = 3$$

$$\text{Maximum value of } x + y = \pi + \frac{5\pi}{2} = \frac{7\pi}{2}$$

$$\text{at } \alpha = 1, \beta = 1$$

$$\Rightarrow \alpha_0 = 1, \beta_0 = 1, \gamma_0 = 7$$

3. Ans (2.00)

$f(x)$ must be constant therefore

$$f(1) = f(2) = f(3)$$

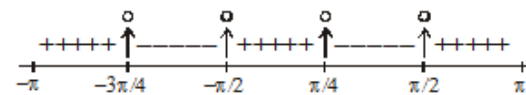
4. Ans (6.00)

$$\text{On solving, } f(x) = e^{2x}$$

$$\therefore g(x) = e^{2x} \cdot \cos^2 x$$

$$\Rightarrow g'(x) = e^{2x} \cdot 2 \cos x (\cos x - \sin x)$$

sign of $g'(x)$



$$g(x) \text{ is local max. at } x = \frac{-3\pi}{4}, \frac{\pi}{4} \Rightarrow n_1 = 2$$

$$g(x) \text{ is local min. at } x = -\frac{\pi}{2}, \frac{\pi}{2} \Rightarrow n_2 = 2$$

$$g(x) \text{ is global min. at } x = \pm \frac{\pi}{2} \Rightarrow n_3 = 2$$

and global min. value of $g(x)$ is zero.

$$\therefore n_1 + n_2 + n_3 = 6.$$

5. Ans (201.00)

$$\frac{dx}{dt} = -2t^{-3} - 6033t^{-4}$$

$$\& \frac{dy}{dt} = -2011t^{-2} - \frac{2}{3}t^{-1}$$

$$\frac{dy}{dx} = \frac{t^2}{3} \Rightarrow 3 \left(\frac{dy}{dx} \right) \left(3x \frac{dy}{dx} - 1 \right)^2 - 1$$

$$= t^2(xt^2 - 1)^2 - 1$$

$$= (2011)^2 - 1 = 2010 \times 2012$$

6. Ans (0.30)

$$y' = \frac{3e^{3\sin^{-1}x}}{\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2} y' = 3y \quad \dots (1)$$

diff. w.r.t-x

$$\sqrt{1-x^2} y'' - \frac{xy'}{\sqrt{1-x^2}} = 3y'$$

$$(1-x^2) y'' - xy' = 3y' \sqrt{1-x^2}$$

$$(1-x^2) y'' - xy' = 9y' (u \sin g (1))$$

diff. w.r.t.x

$$(1-x^2) y''' - 2xy'' - xy'' - y' = 9y'$$

$$(1-x^2) y''' - 3xy'' - 10y' = 0$$

$$(1-x^2) y''' = 3xy'' + 10y'$$

$$\lambda = 3, k = 10$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)
ALLEN COMMON TEST
12-09-2021

JEE(Main+Advanced) : ENTHUSIAST COURSE

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	B	C	D	A	A	A	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	B	A	C	B	D	C	A	D	C
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	40.00	10.00	99.63 to 99.65	50.00	34.37	0.50	5.20	1.50	7.00	5.33

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	A	D	D	D	C	C	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	C	B	A	A	A	D	B	D	D
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	40.00	9.00	82.80	70.58 or 70.59	0.00	13.00	2.25	3.00	4.00	10.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	A	D	B	D	B	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	C	C	B	A	A	D	A	A	B
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	9.00	5.00	1.00	5.00	2.00	4.00	0.00	7.00	4.00	4.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (A)

(A) When m_1 is removed, the unbalanced upward force is $= m_1 g$.

$$\text{So amplitude of motion} = \frac{m_1 g}{k}$$

2. Ans (B)

If Δx_1 and Δx_2 are their respective amplitude of motion, then

$$m\Delta x_1 + 2m(-\Delta x_2) = 0 \quad \dots\dots(i)$$

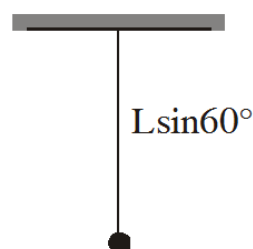
$$\text{and } \Delta x_1 + \Delta x_2 = x_0 \quad \dots\dots(ii)$$

After solving above equations, we get

$$\Delta x_1 = \frac{2x_0}{3}$$

3. Ans (B)

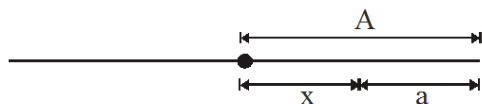
The equivalent pendulum is shown in figure.



$$\therefore T = 2\pi \sqrt{\frac{L \sin 60^\circ}{g}} = 2\pi \sqrt{\frac{\sqrt{3}L}{2g}}$$

4. **Ans (C)**

When particle starts from extreme position,



$$x = A \cos \omega t$$

$$\therefore (A - a) = A \cos \omega t \times 1$$

$$\text{or } (A - a) = A \cos \omega \quad \dots\dots(i)$$

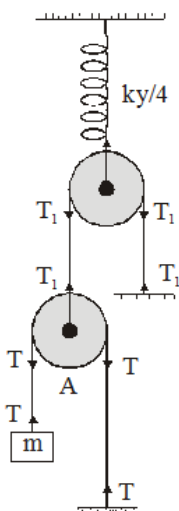
$$\text{Also } A - (a + b) = A \cos \omega \times 2$$

$$\text{or } A - (a + b) = A \cos 2\omega \quad \dots\dots(ii)$$

After simplifying above equations, we get

$$A = \left[\frac{2a^2}{3a - b} \right]$$

5. **Ans (D)**



When one displaces the block by y , the pulley A will go down by $y/2$. Accordingly the pulley B goes down by $y/4$. So spring will stretch by $y/4$.

$$\text{Thus } \frac{ky}{4} = 2T_1$$

$$\text{and } 2T = T_1$$

$$\therefore T = \frac{ky}{16}$$

$$\text{The restoring force, } F = -T = -\frac{k}{16}y$$

$$\text{and acceleration, } a = \frac{F}{m} = \frac{k}{16m}(-y)$$

$$\therefore T = 2\pi\sqrt{\frac{16m}{k}} = 8\pi\sqrt{\frac{m}{k}}$$

6. **Ans (A)**

Earth's gravitation pull is stronger than moon at same distance.

7. **Ans (A)**

$$v_e = \sqrt{\frac{2GM_e}{R_e}} = 11$$

$$\rho_P = \rho_e, R_P = 2R_e$$

$$\frac{M_P}{\frac{4}{3}\pi R_P^3} = \frac{M_e}{\frac{4}{3}\pi R_e^3}$$

$$M_P = M_e \left(\frac{R_P}{R_e} \right)^3 = 8M_e$$

$$\therefore v_P = \sqrt{\frac{2GM_P}{R_P}} = \sqrt{\frac{2 \times 8GM_e}{2R_e}} = \sqrt{4 \times \frac{2GM_e}{R_e}} = 22 \text{ km/s}$$

8. **Ans (A)**

Let v_r is the velocity of approach at the required separation. Then by conservation of mechanical energy. Here by using the concept of reduce mass, we have

$$0 + 0 = \frac{1}{2}\mu v_r^2 - \frac{Gm_1 m_2}{r}$$

$$\text{Here } \mu = \frac{m_1 m_2}{m_1 + m_2}$$

Substituting the value of μ in equation (i) we get

$$v_r = \sqrt{\frac{2G(m_1 + m_2)}{r}}$$

9. **Ans (B)**

$$\text{Given, gravitational field } E_g = (5i + 10j) \text{ N/kg}$$

$$\text{The gravitational force } F = mE_g = 0.4(5i + 10j) = (2i + 4j) \text{ N}$$

$$\text{The displacement } s = (3i + 4j) \text{ m}$$

$$\text{The work done by the gravitational force } W = F \cdot s = (2i + 4j) \cdot (3i + 4j) = 22 \text{ J}$$

$$\text{Thus, the work done by external agent} = -22 \text{ J}$$

10. **Ans (A)**

Here, electrostatic repulsive force = gravitational attractive force

$$\text{i.e., } \frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{r^2} = \frac{GM_2 M_m}{r^2}$$

$$\text{or } q = \sqrt{4\pi\epsilon_0 GM_e M_m}$$

11. **Ans (B)**

We have,

$$\phi_{\text{total}} = \phi_A + \phi_B + \phi_C = \frac{q}{\epsilon_0}$$

$$2\phi' + \phi = \frac{q}{\epsilon_0}$$

$$\Rightarrow \phi' = \frac{1}{2} \left(\frac{q}{\epsilon_0} - \phi \right)$$

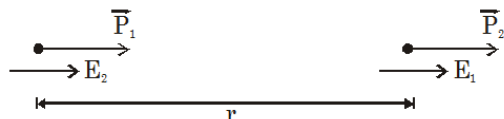
12. **Ans (B)**

In an electric field, electric intensity E and potential V are related to each other through the relation,

$$\vec{E} = -\frac{dV}{dr}\vec{n}, \text{ i.e., } E = -\frac{dV}{dr}$$

And as here for a given line, $V = \text{constant}$, the potential difference between any two consecutive lines $dV = V_1 - V_2 = 10V = \text{const.}$ and hence E will be maximum where the distance dr between the lines is minimum, i.e., at B (where the lines are closest).

13. **Ans (A)**



The P.E. of the dipole $\vec{P}_2$ in the field of $\vec{P}_1$ is

$$U = -P_2 E_1 \cos 0^\circ = -\frac{1}{4\pi\epsilon_0} \frac{2P_1 P_2}{r^3}$$

The force between them is given by

$$F = -\frac{dU}{dr} = -\frac{d}{dr} \left[-\frac{1}{4\pi\epsilon_0} \frac{2P_1 P_2}{r^3} \right] = -\frac{1}{4\pi\epsilon_0} \frac{6P_1 P_2}{r^4}$$

14. **Ans (C)**

(A) Since V varies linearly with x with a rate of $\frac{dV}{dx} = 5$, we have

$$V = 5x + C;$$

where $x = 0$, $V = 5$ or $C = 5$

$$\text{Then, } V_2 = 5x + 5 \dots\dots(i)$$

$$\text{Since, } \tan \theta = \frac{dy}{dx} = \sqrt{3}$$

$$y = \sqrt{3}x \dots\dots(ii)$$

$$\text{Hence } V = \frac{5}{\sqrt{3}}y \dots\dots(iii)$$

Adding Eqs. (i) and (iii),

$$V = 5 \left(x + \frac{y}{\sqrt{3}} \right) + 5$$

15. **Ans (B)**

(a) As in sharing, charge divides in proportion to capacity

$$\frac{q_1}{q_2} = \frac{C_1}{C_2} = \frac{R_1}{R_2} = \frac{R}{2R} = \frac{1}{2}$$

i.e., larger sphere has greater charge.

(b) As charge density, $\sigma = q/(4\pi R^2)$

$$\frac{\sigma_1}{\sigma_2} = \frac{q_1}{q_2} \times \left[\frac{R_2}{R_1} \right]^2 = \frac{1}{2} \times \left[\frac{2R}{R} \right]^2 = 2$$

$$\left[\text{as } \frac{q_1}{q_2} = \frac{1}{2} \right]$$

i.e., smaller sphere has greater charge density.

16. **Ans (D)**

(a) As capacity of a capacitor depends only on its geometry and is independent of charge given or potential applied, it will remain unchanged, i.e., $C' = C$

(b) As, $q' = C' V' = C(2V)$ [as $C' = C$ and $V' = 2V$]

$$\text{i.e., } q' = 2CV = 2q \quad [\text{as } q = CV]$$

i.e., charge on capacitor is doubled

(c) As, $E' = (V'/d) = (2V/d) = 2E$

$$[\text{as } V' = 2V \text{ and } (V/d) = E]$$

i.e., Field (between the plates of capacitor) is doubled

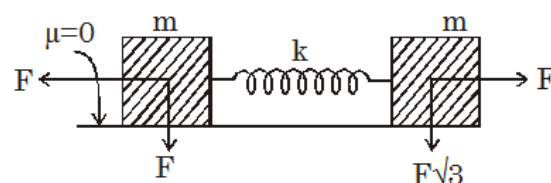
$$(d) \text{ As, } U' = \frac{1}{2} C' (V')^2 = \frac{1}{2} C (2V)^2$$

$$[\text{as } C' = C \text{ and } V' = 2V]$$

$$\text{i.e., } U' = 4 \left[\frac{1}{2} CV^2 \right] = 4U \quad [\text{as } U = \frac{1}{2} CV^2]$$

i.e., energy associated with the capacitor becomes four times of its initial value.

17. **Ans (C)**



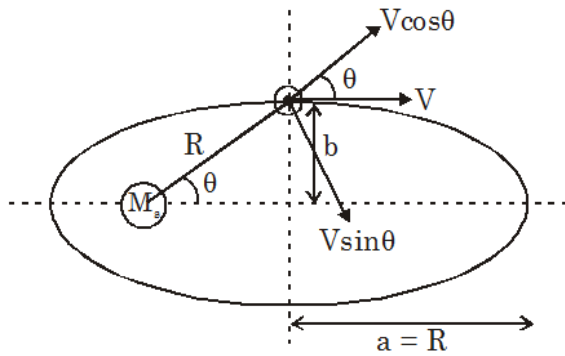
Let maximum elongation be x i.e. $\frac{x}{2} + \frac{x}{2}$

$$\frac{Fx}{2} + \frac{Fx}{2} + \frac{1}{2}k(0^2 - x^2) = 0 - 0$$

$$Fx = \frac{1}{2}kx^2$$

$$\Rightarrow x = \frac{2F}{k}$$

18. Ans (A)



$$T = 2\pi \sqrt{\frac{R^3}{GM_s}}$$

$$\sin \theta = \frac{b}{R}$$

$$\frac{A}{T} = \frac{L}{2m}$$

$$\frac{\pi R b}{2\pi \sqrt{\frac{R^3}{GM_s}}} = \frac{m V \sin \theta R}{2m}$$

$$\Rightarrow V = \sqrt{\frac{GM_s}{R}}$$

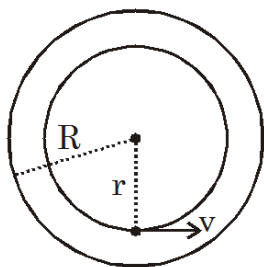
19. Ans (D)

$$|U| = \frac{3}{5} \frac{GM^2}{R}$$

$$M = \rho \cdot \frac{4}{3} \pi R^3$$

$$\Rightarrow |U| \propto R^5$$

20. Ans (C)



Since gravitational field of M is zero inside the cavity, so no gravitational force acts on the particle. So only normal force will act on the particle which provides the centripetal force.

$\Rightarrow$ Speed will remain constant.

$$\Rightarrow T = \frac{\text{distance}}{\text{speed}} = \frac{2\pi r}{v}$$

SECTION-II

1. Ans (40.00)

F_{net} when pressed x distance is

$$F_{\text{net}} = -\rho A x g$$

$$m\omega^2 x = -\rho A g x$$

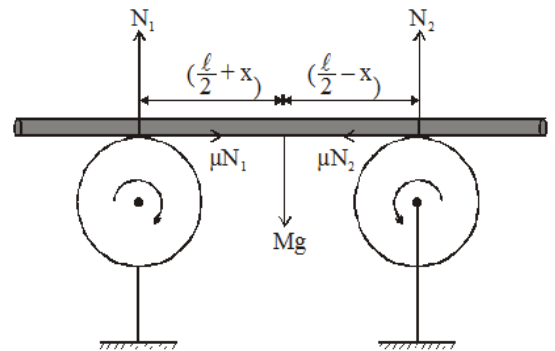
$$\omega^2 = \frac{\rho A g}{M}$$

$$\therefore T = 2\pi \sqrt{\frac{M}{\rho A g}}$$

by solving

$$T = 40 \text{ ms}$$

2. Ans (10.00)



$$N_1 + N_2 = Mg \dots\dots(1)$$

$$\mu N_1 - \mu N_2 = Ma \dots\dots(2)$$

$$N_1 \left(\frac{l}{2} + x \right) = N_2 \left(\frac{l}{2} - x \right) \dots\dots(3)$$

by solving

$$a = - \left(\frac{2\mu g}{l} \right) x$$

$$\therefore \omega^2 = \frac{2\mu g}{l}$$

$$T = 2\pi \sqrt{\frac{l}{2\mu g}}$$

by putting values

$$T = 10 \text{ sec}$$

3. **Ans (99.63 to 99.65)**

Suppose W_0 is the weight of the train when it is not moving, earth is also not rotating. Let W is the weight of the train when earth is rotating with angular velocity ω_e , train is at rest with respect to earth

$$W = W_0 - \frac{mv_e^2}{R}$$

$$\text{or } W_0 = W + \frac{mv_e^2}{R} \quad \text{.....(i)}$$

where

$$\omega_e = \frac{2\pi}{T} = \frac{2\pi}{24 \times 60 \times 60} = 7.27 \times 10^{-5} \text{ rad/s}$$

(i) Apparent weight of the train when it is moving along east, with velocity v w.r.t. earth

$$W' = W_0 - \frac{m(v_e + v)^2}{R}$$

$$= \left(W + \frac{mv_e^2}{R} \right) - \frac{m(v_e + v)^2}{R} \quad \text{.....(ii)}$$

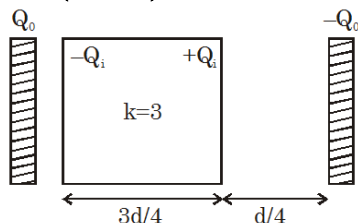
Simplifying equation (ii) and noting that $v \ll v_e$ and $W = mg$, we get

$$W' = W \left(1 - \frac{2\omega_e v}{g} \right)$$

Here $v = 72 \text{ km/h} = 20 \text{ m/s}$ and $\omega_e = 7.27 \times 10^{-5} \text{ rad/s}$

$$\therefore W' = \left(1 - \frac{2 \times 7.27 \times 10^{-5} \times 20}{9.8} \right) = 99.63 \text{ N}$$

4. **Ans (50.00)**



$$\text{Induced charge } Q_i = Q_0 \left(1 - \frac{1}{k} \right) = \frac{2Q_0}{3}$$

$$E_{\text{net inside dielectric}} = \frac{Q_0}{3A\epsilon_0}$$

$\Rightarrow$ Energy stored in region containing dielectric

$$= \frac{1}{2} (k\epsilon_0) E^2 \times \text{volume}$$

$$= \frac{1}{2} (3\epsilon_0) \left(\frac{Q_0}{3A\epsilon_0} \right)^2 \times A \left(\frac{3d}{4} \right)$$

$$= \frac{Q_0^2 d}{8A\epsilon_0}$$

Energy stored in region not containing dielectric

$$= \frac{1}{2} \epsilon_0 \left(\frac{Q_0}{A\epsilon_0} \right)^2 \cdot A \cdot \left(\frac{d}{4} \right)$$

$$= \frac{Q_0^2 d}{8A\epsilon_0}$$

$\Rightarrow$ Half energy is stored in dielectric region.

5. **Ans (34.37)**

If the ball has to just complete the circle then the tension must vanish at the topmost point, i.e. $T_2 = 0$.

From Newton's second law

$$T_2 + mg - \frac{q^2}{4\pi\epsilon_0 \ell^2} = \frac{mv^2}{\ell} \quad \text{.....(i)}$$

At the topmost point, $T_2 = 0$

$$mg - \frac{q^2}{4\pi\epsilon_0 \ell^2} = \frac{mv^2}{\ell} \quad \text{.....(ii)}$$

From energy conservation,

Energy at lowest point = Energy at topmost point

$$\frac{1}{2} mu^2 = \frac{1}{2} mv^2 + mg2\ell \quad \text{.....(iii)}$$

$$v^2 = u^2 - 4g\ell \quad \text{.....(iv)}$$

$$\text{From Eq. (ii), } v^2 = g\ell - \frac{q^2}{4\pi\epsilon_0 m\ell} \quad \text{.....(v)}$$

From Eq. (iv) and (v),

$$u = \sqrt{5g\ell - \frac{q^2}{4\pi\epsilon_0 m\ell}} \\ = \left(\frac{275}{8} \right)^{1/2} = 5.86 \text{ m/s}$$

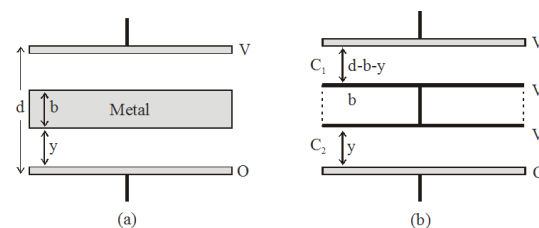
6. **Ans (0.50)**

Before the introduction of the slab the capacity of the system

$$C_0 = \frac{q}{V} = \frac{q}{(Ed)} \quad [\text{as } E = (V/d)]$$

$$\text{or } C_0 = \frac{\sigma A}{(\sigma/\epsilon_0)d} = \frac{\epsilon_0 A}{d} \quad [\text{as } q = \sigma A \text{ and } E = \frac{\sigma}{\epsilon_0}] \quad \text{.....(i)}$$

When the metal slab of thickness b is introduced between the plates, the system becomes equal to two capacitors C_1 and C_2 in series (as metal slab is an equipotential surface) as shown in Fig. (b).



So, the capacity C of the system now becomes

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\text{i.e., } \frac{1}{C} = \frac{(d-b-y)}{\epsilon_0 A} + \frac{y}{\epsilon_0 A} \quad \left[\text{as } C = \frac{\epsilon_0 A}{d} \right]$$

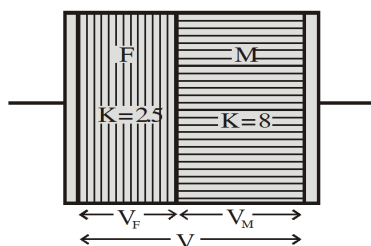
$$\text{or } \frac{1}{C} = \frac{(d-b)}{\epsilon_0 A}, \text{ i.e., } C = \frac{\epsilon_0 A}{(d-b)} \quad \text{.....(ii)}$$

But as here $b = (d/2)$

$$C = \frac{\epsilon_0 A}{(d/2)} = 2 \frac{\epsilon_0 A}{d} = 2C_0, \text{ i.e., } \frac{C}{C_0} = 2$$

7. Ans (5.20)

As the field between the plates of a parallel plate capacitor



$$E = (V/d), \text{ i.e., } V = Ed.$$

$$\text{So } (V_F) = (E_F \times d_F)$$

$$= 6.4 \times 10^6 \times 0.5 \times 10^{-3} = 3.2 \text{ kV.}$$

Now as for a capacitor

$$q = CV ; q_F = C_F V_F \text{ and } q_M = C_M V_M.$$

But here the given capacitor is equivalent to a combination of a fibre and a mica capacitor in series. As in series charge on each capacitor is same.

$$q_M = q_F, C_M V_M = C_F V_F$$

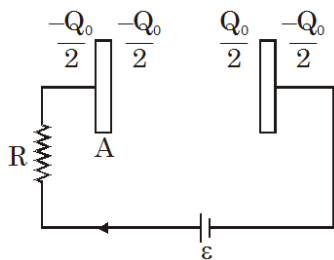
$$\text{or } V_M = \frac{C_F}{C_M} = \left[\frac{K_F}{K_M} \right] \left[\frac{d_M}{d_F} \right] V_F \left[\text{as } C = \frac{\epsilon_0 K A}{d} \right]$$

$$V_M = \left[\frac{2.5}{8} \right] \left[\frac{1 \times 10^{-3}}{0.5 \times 10^{-3}} \right] \times 3.2 \text{ kV} = 2 \text{ kV}$$

$$\text{And as in series } V = V_1 + V_2$$

$$\text{i.e., } V = V_F + V_M = 3.2 \text{ kV} + 2 \text{ kV} = 5.2 \text{ kV}$$

8. Ans (1.50)



Charge flowing from battery at any time t is given by

$$q = C \left(\epsilon + \frac{Q_0}{2C} \right) \left(1 - e^{-\frac{t}{RC}} \right)$$

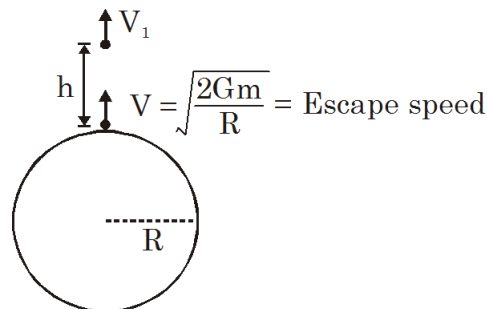
For net charge on plate A to be zero,

$$q = +Q_0$$

$$\Rightarrow Q_0 = C \left(\epsilon + \frac{Q_0}{2C} \right) \left(1 - e^{-\frac{t}{RC}} \right)$$

$$\Rightarrow t = 1.5$$

9. Ans (7.00)



By energy conservation

$$-\frac{GMm}{R+h} + \frac{1}{2}mv_1^2 = -\frac{GMm}{R} + \frac{1}{2}m \left(\sqrt{\frac{2GM}{R}} \right)^2$$

$$\Rightarrow V_1 = \sqrt{\frac{2GM}{R+h}} = \frac{dh}{dt}$$

$$\Rightarrow \sqrt{2GM} \int_0^t dt = \int_0^{3R} \sqrt{R+h} dh$$

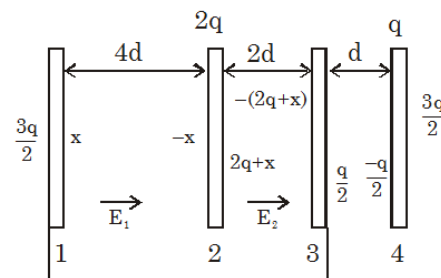
$$\Rightarrow t = \frac{7}{3} \sqrt{\frac{2R^3}{GM}}$$

$$\therefore g = \frac{GM}{R^2} \text{ at earth surface}$$

$$t = \frac{7}{3} \sqrt{\frac{2R}{g}}$$

10. Ans (5.33)

Rearranging the charge



$$V_1 = V_3$$

$$\Rightarrow E_1 (4d) + E_2 (2d) = 0$$

$$\Rightarrow \frac{x}{A\epsilon_0} \cdot 4d + \frac{(2q+x)}{A\epsilon_0} \cdot 2d = 0$$

$$\Rightarrow x = -\frac{2q}{3}$$

$$\Rightarrow \text{charge on right side of plate 2 is } 2q + x = \frac{4q}{3}$$

$$\Rightarrow \Delta V_{23} = \frac{\frac{4q}{3}}{C} = \frac{4q}{3 \cdot \frac{\epsilon_0 A}{2d}}$$

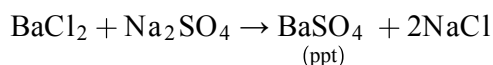
$$\Rightarrow \Delta V_{23} = \frac{8qd}{3\epsilon_0 A} = \frac{8}{3} \times 2$$

$$= \frac{16}{3}$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (C)**



moles $2 \times 0.5 \quad 1 \times 0.5$

taken :

Na_2SO_4 is L.R & BaCl_2 is present in excess.

$$[\text{Ba}^{+2}] = \frac{(n_{\text{Ba}^{+2}})_{\text{left}}}{(500 + 500)} \times 1000$$

$$= (2 \times 0.5 - 1 \times 0.5) = 0.5 \text{ M}$$

2. **Ans (D)**

Solute molecule doesn't pass through SPM.

3. **Ans (A)**

$$P_{\text{Theoretical}} = P_A^0 X_A + P_B^0 X_B$$

$$= \left(200 \times \frac{1}{4} + 160 \times \frac{3}{4} \right)$$

$$= 50 + 120 = 170 \text{ mm of Hg}$$

$$P_{\text{observed}} = 180 \text{ mm Hg} > P_{\text{theoretical}}$$

$\therefore$ solution shows positive deviation from Raoult's law.

4. **Ans (D)**

Theory based.

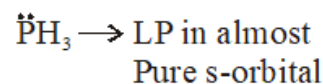
5. **Ans (D)**

Theory based.

6. **Ans (D)**

Theory based : Formula of compound : $A_{1+\delta}B$.

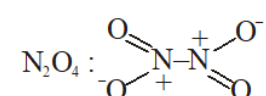
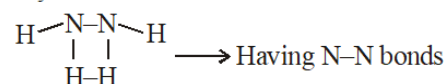
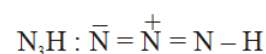
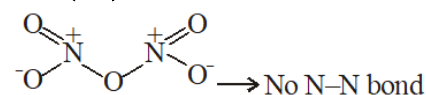
7. **Ans (C)**



9. **Ans (D)**

$$\text{Uut} \xrightarrow[\text{no.}]{\text{at.}} 113$$

10. **Ans (A)**



11. **Ans (B)**

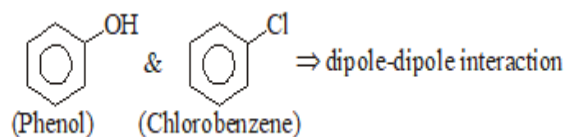
N has more IE_1 than O due to more stable configuration.

12. **Ans (C)**

$[\text{V}(\text{CO})_6]^-$ due to synergic bonding, V-C bond order > 1

13. **Ans (B)**

NaCl & $\text{H}_2\text{O} \Rightarrow$ ion-dipole interaction



option(C) $\Rightarrow P_4 = \text{non-polar}$
option(D) $\Rightarrow \text{CCl}_4 = \text{non-polar}$

SECTION-II

1. **Ans (40.00)**

$$P = P_A^0 + (P_B^0 - P_A^0) X_A$$

$$100 = 120 - 40 X_A$$

$$X_A = \frac{1}{2} \Rightarrow X_B = \frac{1}{2}$$

$$Y_B = \frac{P_B^0 X_B}{P} = \frac{80 \times \frac{1}{2}}{100} = 0.4$$

$$\text{mole \% of B} = Y_B \times 100 = 40$$

2. **Ans (9.00)**

$$300 = \frac{n_1}{v_1} \times R \times 300 \quad \dots\dots(1)$$

$$300 = \frac{n_1}{v_2} \times R \times 327 \quad \dots\dots(2)$$

$$\Rightarrow \frac{v_2}{v_1} = \frac{327}{300}$$

% increases in volume

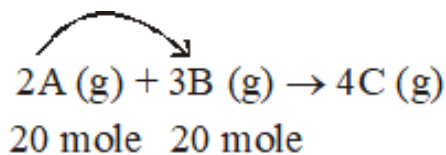
$$= \frac{v_2 - v_1}{v_1} \times 100 = \frac{327 - 300}{300} \times 100 = 9$$

3. **Ans (82.80)**

$$\left(\frac{r_+}{r_-} \right)_{\text{min}^m} = 0.414$$

$$r_+ = 0.414 \times 200 \text{ pm} = 82.8 \text{ pm.}$$

4. Ans (70.58 or 70.59)



$\Rightarrow$ L.R is B

$$\therefore \text{moles of C formed} = \frac{4}{3} \times 20 = \frac{80}{3} \text{ mole}$$

(considering 100% yield)

$\therefore$ Actual number of moles of

$$C = \frac{80}{3} \times \frac{90}{100} = 24 \text{ moles.}$$

$$\text{moles of B unreacted} = 20 \times \frac{10}{100} = 2 \text{ mole}$$

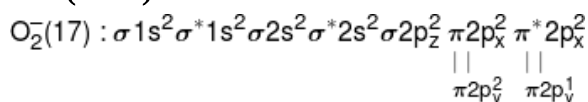
$$\text{moles of A unreacted} = 20 - \frac{2}{3} \times 18 = 8 \text{ mole}$$

$$\%C(\text{by mole}) = \frac{24}{24 + 2 + 8} \times 100$$

$$= 70.588$$

$$\simeq 70.59$$

5. Ans (0.00)



6. Ans (13.00)

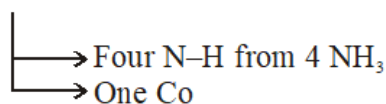
Aluminium (at. number = 13)

7. Ans (2.25)

$$x = 2 (d_{z^2} \text{ and } d_{x^2-y^2})$$

$$p = 2 (\text{cis and trans})$$

$$q = 9 (\text{max. atom in same plane} \rightarrow \text{in trans form})$$



PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$f(x) = (x^2 - 1)(\pi - e^x), f'(x) = 2x(\pi - e^x) - (x^2 - 1)e^x$$

$$f'(-1) < 0 \text{ however } f'(0) > 0.$$

2. Ans (C)

$$\ell n y = (\tan x)^{\tan x}. \ell n (\tan x), y = 1 \text{ when } x = \frac{\pi}{4}$$

$$\Rightarrow \frac{1}{y} y' = (\tan x)^{\tan x} \cdot \frac{1}{\tan x} \cdot \sec^2 x + (\tan x)^{\tan x} \times (1 + \ln (\tan x)) \times \sec^2 x \times \ln (\tan x)$$

$$\text{put } x = \frac{\pi}{4}; y' = 2.$$

3. Ans (B)

$$\lim_{x \rightarrow 0^+} \frac{e^{e/x} - e^{-e/x}}{e^{1/x} + e^{-1/x}}$$

$$= \lim_{x \rightarrow 0^+} \frac{e^{\frac{e-1}{x}} (1 - e^{-2e/x})}{(1 + e^{-2/x})} = +\infty$$

$$\lim_{x \rightarrow 0^-} \frac{e^{e/x} - e^{-e/x}}{e^{1/x} + e^{-1/x}} = \lim_{x \rightarrow 0^-} \frac{e^{-e/x} (e^{2e/x} - 1)}{e^{-1/x} (e^{2/x} + 1)}$$

$$= \lim_{x \rightarrow 0^-} e^{-\left(\frac{e-1}{x}\right)} \left(\frac{e^{2e/x} - 1}{e^{2/x} + 1} \right) = -\infty$$

limit doesn't exist so f(x) is discontinuous

4. Ans (A)

$$\text{Let } 2^{32} = n$$

$$k = \lim_{x \rightarrow 1} \frac{x^n - nx + n - 1}{(x - 1)^2} - \lim_{x \rightarrow 1} \frac{x^n - 1 - n(x - 1)}{(x - 1)^2}$$

$$= \lim_{x \rightarrow 1} \frac{(x^{n-1} + x^{n-2} + \dots + 1) - n}{x - 1}$$

$$= (n - 1) + (n - 2) + \dots + 1 = \frac{2^{32} (2^{32} - 1)}{2}$$

$$= 2^{63} - 2^{31}$$

5. Ans (D)

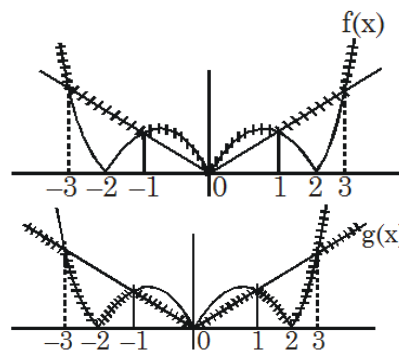
$$f(x) = -x + x - 1, x < 0$$

$$= x - x - 1, x \geq 0$$

f(x) = -1 = a constant function, Hence, differentiable for all x

6. Ans (B)

f(x) is non-differentiable at x = -3, -1, 0, 1, 3 and g(x) is non-differentiable at x = -3, -2, -1, 0, 1, 2, 3 $\Rightarrow$ (B)



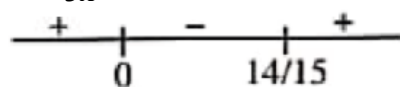
7. **Ans (D)**

$$f(x) = (3x-7)x^{2/3}$$

$$\Rightarrow f(x) = 3x^{5/3} - 7x^{2/3}$$

$$\Rightarrow f'(x) = 5x^{2/3} - \frac{14}{3x^{1/3}}$$

$$= \frac{15x - 14}{3x^{1/3}} > 0$$



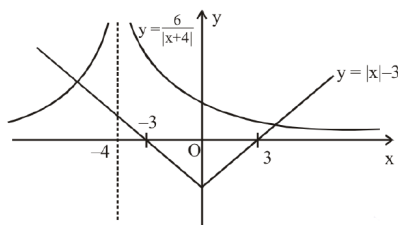
$$\therefore f'(x) \geq 0 \forall x \in (-\infty, 0] \cup \left[\frac{14}{15}, \infty\right)$$

8. **Ans (B)**

$$x \neq -4$$

$$(|x| - 3)(|x + 4|) = 6$$

$$\Rightarrow |x| - 3 = \frac{6}{|x + 4|}$$



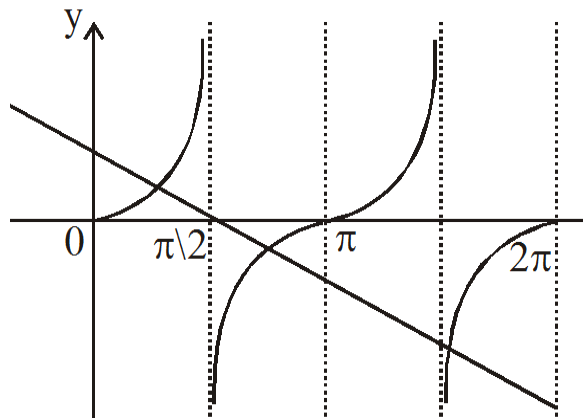
No. of solutions = 2

9. **Ans (A)**

$$x + 2 \tan x = \frac{\pi}{2}$$

$$\Rightarrow 2 \tan x = \frac{\pi}{2} - x$$

$$\Rightarrow \tan x = -\frac{1}{2}x + \frac{\pi}{4}$$



10. **Ans (B)**

$$f(x) = \frac{\cos \operatorname{arccot} x}{\sqrt{|x|}}$$

$$\text{Domain} \in (-\infty, -1] \cup [1, \infty)$$

$$\{x\} \neq 0 \text{ so } x \neq \text{integers}$$

11. **Ans (B)**

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{7 \left[\left(1 + \frac{x}{2401} \right)^{1/4} - 1 \right]}{49 \left(1 - \left(1 + \frac{x}{2401} \right)^{1/2} \right)} \\ &= \lim_{x \rightarrow 0} \frac{-1}{7} \frac{\left(\left(1 + \frac{x}{2401} \right)^{1/4} - 1 \right) \left(1 - \left(1 + \frac{x}{2401} \right)^{1/2} \right)}{\left(1 + \frac{x}{2401} - 1 \right) \left(1 - \left(1 + \frac{x}{2401} \right)^{1/2} \right)} \\ &= \frac{-1}{7} \cdot \frac{1}{4} \cdot \frac{1}{1/2} \\ &= \frac{-1}{14} \end{aligned}$$

12. **Ans (C)**

$$\frac{dy}{dx} = 1 - \sin y \Rightarrow \frac{dy}{dx} = \frac{1}{1 + \sin y}$$

$$\text{Slope of tangent at } (a, b) = \frac{1}{1 + \sin b} = \frac{3/2 - 2}{0 - 1/2}$$

$$\Rightarrow \sin b = 0$$

$$b = n\pi : n \in \mathbb{I} \dots \dots \dots (i)$$

$$\text{and } b = a + \cos b \Rightarrow a = b \pm 1 \dots \dots \dots (ii)$$

13. **Ans (C)**

$f(x+5) = f(x) \Rightarrow f(x)$ is periodic with period 5.

$$\text{Now } f(11) = f(10+1) = f(1) = 1; f(-11) =$$

$$f(-11+15) = f(4) = 4 + 2 = 6$$

$$\therefore \frac{f(11) - f(-11)}{f(11) + f(-11)} = \frac{1 - 6}{1 + 6} = \frac{-5}{7}$$

14. **Ans (B)**

$$\begin{aligned} & \tan^{-1} \left(\frac{4}{1+2.6} \right) + \tan^{-1} \left(\frac{10}{1+3.8} \right) + \tan^{-1} \left(\frac{18}{1+4.10} \right) \dots \infty \\ &= \tan^{-1} \left(\frac{6-2}{1+2.(6)} \right) + \tan^{-1} \left(\frac{12-2}{1+2.12} \right) + \tan^{-1} \left(\frac{20-2}{1+2.20} \right) \dots \infty \\ &= \lim_{n \rightarrow \infty} \left(\tan^{-1} (n+1)(n+2) - \tan^{-1} 2 \right) = \frac{\pi}{2} - \tan^{-1} 2 = \cot^{-1} 2 \end{aligned}$$

15. **Ans (A)**

$$\begin{aligned} & \tan^{-1} \left(\frac{\frac{1}{5} + \frac{1}{7}}{1 - \frac{1}{5} \times \frac{1}{7}} \right) + \tan^{-1} \left(\frac{\frac{1}{3} + \frac{1}{8}}{1 - \frac{1}{3} \times \frac{1}{8}} \right) = \tan^{-1} \left(\frac{6}{17} \right) + \tan^{-1} \left(\frac{11}{23} \right) \\ &= \tan^{-1} \left(\frac{\frac{6}{17} + \frac{11}{23}}{1 - \frac{6}{17} \times \frac{11}{23}} \right) = \tan^{-1} \left(\frac{325}{325} \right) = \tan^{-1} (1) = \frac{\pi}{4} \end{aligned}$$

16. Ans (A)

$$kx + y = e^{xy} \text{ crosses x-axis at } \left(\frac{1}{k}, 0\right)$$

$$k + y' = e^{xy}(xy' + y)$$

$$x = \frac{1}{k}, y = 0 \Rightarrow y' = \frac{k^2}{1-k}$$

$$\text{Normal : } y - 0 = \frac{k-1}{k^2} \left(x - \frac{1}{k}\right)$$

$$\text{Area (A)} = \frac{1}{2} \left(\frac{k-1}{k^2}\right)$$

$$\frac{dA}{dk} = \frac{1}{2} \left(-\frac{3}{k^4} + \frac{4}{k^5}\right)$$

$$\Rightarrow k = \frac{4}{3}$$

17. Ans (D)

$$S = 16t + 11t^2 - t^3$$

$$\text{Velocity (v)} = \frac{dS}{dt} = 16 + 22t - 3t^2$$

$$\frac{dS}{dt} = 0 \Rightarrow (3t+2)(t-8) = 0$$

$$\Rightarrow t = 8 \text{ seconds}$$

18. Ans (A)

$f(x)$ is continuous and differentiable

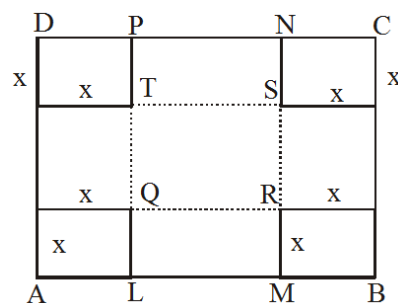
$$f(1) = f(2) = 0$$

$$\text{so } f'(c) = 0, c \in (1, 2)$$

$$(c-2)^2 + (c-1) \cdot 2(c-2) = 0$$

$$c = 4/3$$

19. Ans (A)



Let x be the length of each side of the square cut out

$$\text{Volume of the box} = V = x(6-2x)^2$$

$$\Rightarrow \frac{dV}{dx} = (6-2x)^2 + x \cdot 2(6-2x)(-2)$$

$$\Rightarrow \frac{dV}{dx} = (6-2x)(6-2x-4x)$$

$$\Rightarrow \frac{dV}{dx} = 12(3-x)(1-x)$$

$$\Rightarrow \frac{d^2V}{dx^2} = -24(2-x)$$

$$\text{For max or min } \frac{dV}{dx} = 0 \text{ gives}$$

$$\Rightarrow x = 3, 1$$

$$\text{At } x = 1, \frac{d^2V}{dx^2} = -24 < 0$$

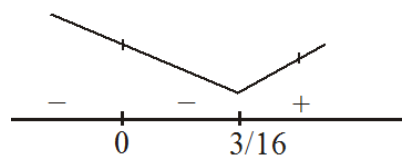
So the volume is maximum

Hence the height of the box = 1 ft.

20. Ans (B)

$$f'(x) = 16x^3 - 3x^2 = 0 \Rightarrow x^2(16x-3)$$

$$f''(x) = 48x^2 - 6x = x(48x-6)$$



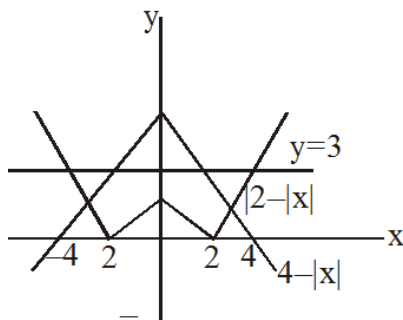
Concavity changer at $x=0$; 50 point is (0, 2)

SECTION-II

1. Ans (9.00)

$$\begin{aligned} M &= \lim_{x \rightarrow 0} \left(\frac{1 - (f(x))^3}{5 \tan^2 x} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{(1 - f(x)) (1 + f(x) + f^2(x))}{5 \left(\frac{\tan^2 x}{x^2} \right) x^2} \right) \\ &= \lim_{x \rightarrow 0} \left(\frac{3(1 - f(x))}{5x^2} \right) \\ &= \frac{3}{5} \lim_{x \rightarrow 0} \left(\frac{1 - \cos 2x \cos 4x \cos 6x \cos 8x \cos 10x}{x^2} \right) \\ &= \frac{3}{5} \left(\frac{2^2 + 4^2 + 6^2 + 8^2 + 10^2}{2} \right) \\ &= \frac{3}{5} \times 2^2 \left(\frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2}{2} \right) \\ &= \frac{12}{5} \times \frac{5.611}{12} = 66 \end{aligned}$$

2. Ans (5.00)



3. Ans (1.00)

$$\begin{aligned} L &= \lim_{x \rightarrow 0} \frac{\tan(\pi e^{x^2} - \pi)}{\sin^2\left(\frac{\pi}{2} - \frac{\pi}{2} e^x\right)} = \lim_{x \rightarrow 0} \frac{\pi(e^{x^2} - 1)}{\sin^2\left(\frac{\pi}{2}(1 - e^x)\right)^2} \\ &= \lim_{x \rightarrow 0} \frac{4}{\pi} \frac{e^{x^2} - 1}{(e^x - 1)} = \frac{4}{\pi} \end{aligned}$$

$$[L] = 1$$

4. Ans (5.00)

$$\begin{aligned} \frac{dy}{dx} &= \frac{3px^2}{2y} \\ m &= \left(\frac{dy}{dx} \right)_{(2,3)} = \frac{3px^2}{2y} = \frac{12p}{6} = 2p \end{aligned}$$

Hence, the equation of the tangent at (2, 3) is

$$y - 3 = 2p(x - 2)$$

which is identical with $y = 4x - 5$

$$\text{Thus, } 2p = 4 \Rightarrow p = 2$$

Since the point (2, 3) lies on the curve, so

$$8p + q = 9$$

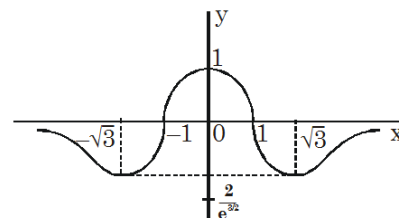
$$\Rightarrow 16 + q = 9$$

$$\Rightarrow q = 9 - 16 = -7$$

5. Ans (2.00)

$$\text{Let } f(x) = e^{-x^2/2}(1 - x^2)$$

$$f'(x) = e^{-x^2/2}(-2x - x(1 - x^2)) = e^{-x^2/2} \times (x^2 - 3)$$



$$\Rightarrow k = 0, 1$$

6. Ans (4.00)

Use LMVT for $f(x)$ in $[1, 4]$

$$f'(c_1) = \frac{f(4) - f(1)}{4 - 1} \geq 2$$

$$\Rightarrow f(4) \geq 4$$

7. Ans (0.00)

$f(x)$ is a cubic with leading co-efficient positive, so can not be always decreasing for $x > 0$

8. **Ans (7.00)**

Since $f: [1, 7] \rightarrow \mathbb{Q}$ be a continuous function where $f(1) = 7$, so it is a constant function.

Thus, $f(7) = 7$.

9. **Ans (4.00)**

Given $F(x^2) = x^2 + x^3$

$$\Rightarrow F'(x^2) \cdot 2x = 2x + 3x^2$$

$$\Rightarrow F'(x^2) = \frac{2x + 3x^2}{2x} = 1 + \frac{3}{2}x$$

$$\Rightarrow f(x^2) = 1 + \frac{3}{2}x$$

Put $x = 2$

$$\Rightarrow f(4) = 1 + \frac{3}{2} \cdot 2 = 1 + 3 = 4$$

10. **Ans (4.00)**

Given $f(x) = x^2 + x^3$

As we know that, if g is the inverse of f , then

$$y = f(x) \Leftrightarrow x = g(y)$$

$$\text{Now, } \frac{dy}{dx} = 2x + 3x^2$$

$$\Rightarrow g'(y) = \frac{dx}{dy} = \frac{1}{2x + 3x^2}$$

when $y = 2$, then $x = 1$

$$\text{Now, } g'(2) = \left(\frac{dx}{dy} \right)_{\substack{x=1 \\ y=2}} = \frac{1}{2+3} = \frac{1}{5}$$

Hence, the value of $(5g'(2) + 3)$

$$= 1 + 3$$

$$= 4$$